Modern CLI Tools for Sysadmins

A Curated Collection of Tools to Transform Your Workflow

From the dev-env Project

What We'll Cover Today

- **Foundation**: Zsh & Zellij - Modern shell and terminal management
- **Navigation**: File finding, fuzzy search, and smart directory jumping
- **Search & Text**: Fast search and beautiful file viewing
- **Monitoring**: Process, system, network, and I/O monitoring
- **Version Control**: Terminal UIs for git
- **Containers**: Podman, distrobox, and container management
- **Development**: Editors, database clients, API tools
- **Utilities**: Helpers that make life easier

All installed via one Ansible playbook - reproducible and version-controlled

Part 1: Foundation

Z Shell & Oh-My-Zsh

Z Shell (Zsh) + Oh-My-Zsh

What is it?

Modern, feature-rich shell extending bash with powerful auto-completion, better globbing, and extensive customization. Oh-My-Zsh adds 200+ plugins and easy configuration.

Why adopt it?

- ✅ Command correction - typo detection and suggestions
- ✅ Advanced tab completion - context-aware for commands, flags, paths
- ✅ Shared history across all terminal sessions
- ✅ Plugin ecosystem - git status, AWS, kubectl, and more

Zsh: Key Features to Demo

1. Intelligent Auto-completion

```
kill -<TAB>           # Shows all signal options
cd /u/l/b<TAB>        # Expands to /usr/local/bin
```

2. Command History Search

```
Ctrl+R                # Reverse history search with fzf
# Type 'ssh' then ↑   # Shows only ssh commands
```

3. Plugins Included

- `zsh-autosuggestions` - Ghost text from history
- `zsh-syntax-highlighting` - Real-time validation (red/green)
- `zsh-fzf-history-search` - Fuzzy search with Ctrl+R

Zsh: Spaceship Prompt

Shows critical info at a glance:

```
~/project main ⇡1 !2 [+3 ~1 -1]  # Git branch, unpushed commits, changes
took 2s                           # Execution time for long commands
```

- Git branch and status automatically displayed
- Shows Python/Node versions when in project directories
- Execution time for commands > 1 second
- Customizable and beautiful

Zellij

Modern Terminal Multiplexer

Zellij - Terminal Workspace Manager

What is it?

Modern terminal multiplexer (like tmux) with better UX, tabs, panes, layouts, and session persistence.

Why adopt it?

- ✅ Session persistence - survive disconnects, perfect for SSH
- ✅ No memorizing keybindings - on-screen hints
- ✅ Built-in layouts - quickly split for monitoring
- ✅ Better than screen/tmux - modern defaults, easier config

Zellij: Essential Commands

Session Management

```
zellij                    # Start new session
zellij attach             # Reconnect to session
```

Pane Management (Ctrl+P)

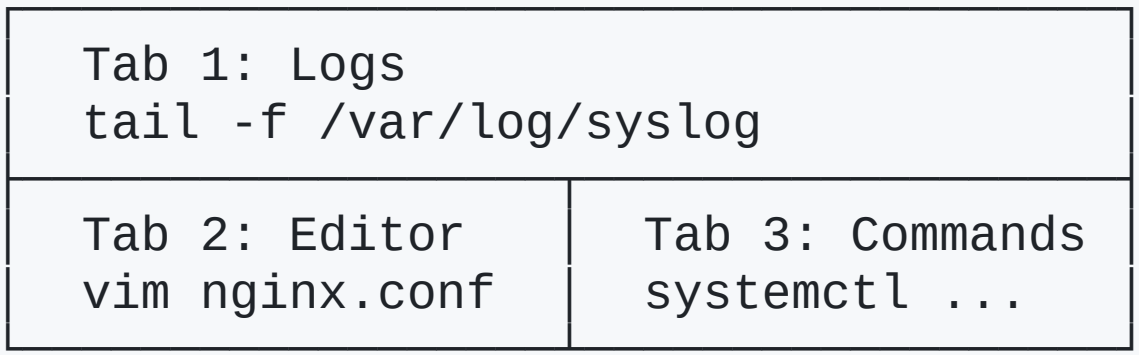
- Split horizontally and vertically
- Resize with shortcuts shown on screen
- Close panes easily

Tab Management (Ctrl+T)

- Create tabs for different tasks
- Rename tabs for organization

Zellij: Real-World Use Case

Demo Layout:



Disconnect → SSH back in → Still there!

Part 2: File & Directory Navigation

FZF - Fuzzy Finder

What is it?

Interactive command-line fuzzy finder. Filters files, history, processes in real-time.

Why adopt it?

- ✅ Find files without exact names
- ✅ Search command history interactively
- ✅ Integrates with git, vim, cd

FZF: Key Bindings

```
Ctrl+R          # Search command history (fuzzy)
Ctrl+T          # Find file, insert path into command
Alt+C           # Fuzzy find directory and cd

# Pipe anything to fzf
ls /var/log | fzf
docker ps -a | fzf
```

Example:

Type "ngnx" → matches "nginx", "nginx.conf", etc.

Zoxide - Smart CD

What is it?

Smarter cd that learns your most-visited directories. Jump with minimal typing.

Why adopt it?

- ✅ No more `cd ../../var/log/nginx` repeatedly
- ✅ Jump to frequent dirs: `z ngi` → `/var/log/nginx`
- ✅ Learns your habits, ranks by frequency

Commands:

```
z nginx             # Jump to most frequent nginx directory
zi                  # Interactive selection with fzf
zoxide query        # See ranked directories
```

Eza & LSD - Modern LS

What are they?

Modern `ls` replacements with colors, icons, git integration, and tree views.

Why adopt them?

- ✅ Git integration - see modified/staged files
- ✅ Color-coded permissions
- ✅ Tree view built-in
- ✅ Icons for file types

```
eza -la                    # Detailed listing with icons
eza --git                  # Show git status
eza --tree --level=2       # Directory tree
```

FD - Fast Find Alternative

What is it?

User-friendly `find` alternative. Fast, simple syntax, respects `.gitignore`.

Why adopt it?

- ✅ 10x faster than find
- ✅ Simple syntax: `fd pattern`
- ✅ Ignores .git, node_modules automatically
- ✅ Colorized output

Examples:

```
fd nginx                    # vs find . -name "*nginx*"
fd -e conf                  # Find all .conf files
fd -t d lib                 # Find directories named 'lib'
fd -H .env                  # Include hidden files
```

Yazi - Terminal File Manager

What is it?

Fast terminal file manager with vim keybindings, preview, and bulk operations.

Why adopt it?

- ✅ Visual file browsing
- ✅ Preview text, images, PDFs
- ✅ Bulk operations (copy/move/delete)
- ✅ Keyboard-driven, no mouse needed

Demo:

- Launch with `yazi`
- Navigate with hjkl or arrows
- Preview files in panel

Broot - Tree View & Navigator

What is it?

Interactive directory navigator with tree views, file sizes, fuzzy search for large directories.

Why adopt it?

- ✅ Handles huge directories without hanging
- ✅ Combined tree + du functionality
- ✅ Fuzzy search to filter tree
- ✅ Quick navigation and cd

Commands:

```
br                  # Launch broot
# Type to filter tree in real-time
```

Part 3: Search & Text Processing

Ripgrep (rg) - Ultra-Fast Search

What is it?

Blazing fast recursive search. Respects `.gitignore`. Replaces grep/ag/ack.

Why adopt it?

- ✅ 10-100x faster than grep on large codebases
- ✅ Skips .git, binaries, respects .gitignore automatically
- ✅ Colored output with line numbers
- ✅ Simple syntax, better defaults

Examples:

```
rg "error" /var/log             # Search logs for "error"
rg -i "failed"                  # Case-insensitive
rg -t py "import"               # Search only Python files
rg -A 3 -B 3 "error"            # Show 3 lines context
```

Bat - Cat with Superpowers

What is it?

`cat` clone with syntax highlighting, git integration, line numbers, and paging.

Why adopt it?

- ✅ Syntax highlighting for code/configs
- ✅ Git integration - shows changes
- ✅ Line numbers for reference
- ✅ Automatic paging (no more `| less`)

Examples:

```
bat /etc/nginx/nginx.conf                # Syntax highlighted
bat --style=numbers,changes script.py    # Show git changes
bat --paging=never file.txt              # Quick view, no pager
```

Hexyl - Hex Viewer

What is it?

Beautiful hex viewer with colored output distinguishing ASCII, null bytes, binary data.

Why adopt it?

- ✅ Inspect binary files easily
- ✅ Colored output for data types
- ✅ Side-by-side hex + ASCII view

```
hexyl /bin/ls | head -50          # View binary
hexyl suspicious-file.bin         # Examine unknown files
```

UP - Ultimate Plumber

What is it?

Interactive pipe-building tool. Write complex commands interactively, see results in real-time.

Why adopt it?

- ✅ See pipe results as you type
- ✅ Debug pipeline failures
- ✅ Build complex grep/awk/sed chains iteratively

Demo:

```
up /var/log/syslog
# Interactively build: grep error | grep -v INFO | awk '{print $1, $5}'
# Copy final command when done
```

Part 4: Process & System Monitoring

Procs - Modern Process Viewer

What is it?

Modern `ps` replacement with colors, tree view, better default columns.

Why adopt it?

- ✅ Human-readable output with colors
- ✅ Tree view of parent-child processes
- ✅ Sensible defaults without complex flags
- ✅ Built-in search/filter

Examples:

```
procs                    # All processes, colorized
procs nginx              # Search for nginx
procs --tree             # Process tree view
```

Bottom - System Monitor

What is it?

Cross-platform graphical system monitor for terminal. htop + iotop + nethogs combined.

Why adopt it?

- ✅ All-in-one: CPU, memory, disk, network, processes
- ✅ Graphs and charts in terminal
- ✅ Customizable views and layouts
- ✅ Works over SSH

Launch:

```
btm          # Full dashboard with graphs
```

Show CPU/memory graphs, process view, network usage, kill processes interactively.

Nmon - Performance Monitor

What is it?

System performance monitoring tool showing CPU, memory, disk I/O, network in one dashboard.

Why adopt it?

- ✅ Comprehensive view of all stats
- ✅ Interactive - toggle views with keys
- ✅ Recording mode for later analysis
- ✅ Perfect for troubleshooting incidents

Usage:

```
nmon                                # Interactive mode
# Press: c (CPU), m (memory), d (disk), n (network)
```

Iotop - I/O Monitoring

What is it?

Shows I/O usage by processes. Like top, but for disk operations.

Why adopt it?

- ✅ Find I/O hogs instantly
- ✅ Real-time disk reads/writes
- ✅ Sort by I/O activity

Usage:

```
sudo iotop                    # Real-time disk I/O
# Sort by write (w), read (r)
```

Use case: Database slow? Check if logs are hammering the disk.

Iftop - Network Bandwidth Monitor

What is it?

Shows network bandwidth usage by connection. See which IPs/hosts consume bandwidth.

Why adopt it?

- ✅ Find bandwidth hogs instantly
- ✅ Track active connections and speeds
- ✅ Per-host statistics

Usage:

```
sudo iftop -i eth0        # Monitor interface
# Press 't' to toggle views
```


Use case: Bandwidth spike? See exactly which connection is responsible

Gping - Graphical Ping

What is it?

Ping with a graph. Visualizes ping times over time.

Why adopt it?

- ✅ Visual debugging of network latency
- ✅ Ping multiple hosts simultaneously
- ✅ Spot intermittent issues easily

Examples:

```
gping 8.8.8.8                          # Single host graph
gping 8.8.8.8 1.1.1.1 google.com       # Compare multiple
```

Latency spikes appear as graph peaks - impossible to miss!

Bandwhich - Bandwidth by Process

What is it?

Shows which processes use network bandwidth, with connection details.

Why adopt it?

- ✅ Process-level network visibility
- ✅ Real-time monitoring
- ✅ See remote IPs, ports, protocols

Usage:

```
sudo bandwhich              # Launch bandwidth monitor
```

Use case: Mystery bandwidth usage? Find the exact process responsible.

Part 5: Version Control Tools

GitUI

What is it?

Blazing fast terminal UI for git (Rust). Keyboard-driven for commits, staging, branching, history.

Why adopt it?

- ✅ Speed - faster than lazygit, handles large repos
- ✅ Keyboard-centric - no mouse, vim-like
- ✅ Visual workflow - see changes, stage hunks
- ✅ Async operations - background fetch

Demo:

```
gitui                    # Launch in git repo
```

Lazygit

What is it?

Simple terminal UI for git. More beginner-friendly than gitui.

Why adopt it?

- ✅ Clear labels and help text
- ✅ Visual branch structure
- ✅ Interactive rebase - squash, reword, reorder
- ✅ Conflict resolution UI

Demo:

```
lazygit                    # Launch in git repo
```

Comparison: GitUI for speed and large repos, Lazygit for learning and visual clarity.

Part 6: Containers & Virtualization

Podman

What is it?

Daemonless container engine compatible with Docker. Rootless by default, more secure.

Why adopt it?

- ✅ No daemon - no single point of failure
- ✅ Rootless containers - better security
- ✅ Docker compatible - same commands, images
- ✅ Systemd integration - generate units from containers

Examples:

```
podman run -it ubuntu bash          # Run container
podman ps                           # List containers
```

Podman-TUI

What is it?

Terminal UI for Podman. Manage containers, images, volumes, networks visually.

Why adopt it?

- ✅ See all resources at a glance
- ✅ Bulk operations on containers
- ✅ View logs and stats inline
- ✅ Faster than typing CLI commands

Launch:

```
podman-tui                    # Interactive container management
```

Navigate tabs: containers, images, volumes. Start/stop, view logs, inspect details.

Distrobox

What is it?

Run any Linux distribution inside containers with host integration. Home, users, devices shared.

Why adopt it?

- ✅ Test on different distros (Fedora on Ubuntu, etc.)
- ✅ Use packages from any distro's repos
- ✅ Isolated dev environments per project
- ✅ Seamless - apps run as if native

Examples:

```
distrobox create --name fedora --image fedora:latest
distrobox enter fedora                    # Your home dir is shared!
```

Part 7: Development & Productivity

Helix - Modal Text Editor

What is it?

Post-modern modal editor with built-in LSP, multiple cursors, tree-sitter parsing.

Why adopt it?

- ✅ Zero configuration - LSP works out of box
- ✅ Multiple cursors - edit multiple locations
- ✅ Built-in LSP - completions, diagnostics, go-to-def
- ✅ Modern architecture - accurate syntax

Usage:

```
hx /etc/nginx/nginx.conf              # Open file
# Multiple cursors: C (add cursor below)
# LSP features work automatically
```

Lazysql - Database TUI

What is it?

Terminal UI for SQL databases (MySQL, PostgreSQL, SQLite). Browse tables, run queries visually.

Why adopt it?

- ✅ Visual query building
- ✅ Works over SSH, no GUI needed
- ✅ Manage multiple databases
- ✅ Safe - confirms destructive queries

Launch:

```
lazysql                    # Connect to database
```

Slumber - API Client

What is it?

Terminal-based HTTP/REST API client. Postman/Insomnia for the terminal.

Why adopt it?

- ✅ Test APIs without GUI, works over SSH
- ✅ Save request collections
- ✅ Environment variables (dev/staging/prod)
- ✅ Response syntax highlighting

Demo:

```
slumber                    # Launch client
```

Create GET request, show response, save to collection, demonstrate headers/params.

NVM - Node Version Manager

What is it?

Manage multiple Node.js versions on same system. Switch per project.

Why adopt it?

- ✅ Test on different Node versions
- ✅ Per-project versions (.nvmrc)
- ✅ Easy switching: `nvm use 18`
- ✅ No sudo for global packages

Commands:

```
nvm list                    # Installed versions
nvm install 20              # Install Node 20
nvm use 18                  # Switch to Node 18
node --version              # Verify
```

Part 8: Utilities & Helpers

TLDR - Simplified Man Pages

What is it?

Community-driven man pages with practical examples. Skip to examples immediately.

Why adopt it?

- ✅ Quick reference with examples
- ✅ Learn by example, not specs
- ✅ 1000+ commands covered
- ✅ Faster than googling

Examples:

```
tldr tar                    # vs man tar (pages of text)
tldr rsync                  # Common rsync use cases
tldr curl                   # HTTP request examples
```

Navi - Interactive Cheatsheets

What is it?

Interactive cheatsheet tool. Search commands, fill placeholders interactively.

Why adopt it?

- ✅ Command templates with placeholders
- ✅ Interactive variable prompts
- ✅ Community cheatsheets available
- ✅ Create team-specific references

Demo:

```
navi                        # Launch cheatsheet browser
# Search "ssh tunnel"
# Fill in variables interactively
# Execute or copy command
```

Dust - Disk Usage Analyzer

What is it?

Modern `du` that visualizes disk usage in a tree. Find space hogs instantly.

Why adopt it?

- ✅ Visual tree of directory sizes
- ✅ Sorted by size (largest first)
- ✅ Faster than du (parallel scanning)
- ✅ Color-coded for visibility

Examples:

```
dust /var/log              # Analyze directory
dust -d 3                  # Limit depth for quick overview
```

XH - HTTP Client

What is it?

Friendly HTTP client with expressive syntax. Like HTTPie but faster (Rust).

Why adopt it?

- ✅ Human-friendly syntax (simpler than curl)
- ✅ Automatic JSON serialization
- ✅ Colored output
- ✅ Download progress bars

Examples:

```
xh httpbin.org/get                          # Simple GET
xh POST httpbin.org/post name=admin         # POST with JSON
xh -d httpbin.org/image/png > image.png     # Download file
```

Espanso - Text Expander

What is it?

Cross-platform text expander. Type shortcuts that expand to longer text.

Why adopt it?

- ✅ Time saver for common commands/emails
- ✅ Ensure consistent templates
- ✅ Works everywhere (terminal, browser, apps)
- ✅ Dynamic expansions (date, command output)

Examples:

```
:email     → Full email signature
:date      → 2026-01-22
:sshprod   → ssh -i ~/.ssh/prod.pem user@prod-server.com
```

Watchexec - File Watcher

What is it?

Executes commands when files change. Monitor configs, auto-restart services.

Why adopt it?

- ✅ Auto-restart services on config changes
- ✅ Run tests automatically on save
- ✅ Trigger alerts on log patterns
- ✅ Simple syntax vs inotify-tools

Examples:

```
watchexec -w /etc/nginx/nginx.conf -- nginx -t    # Test on change
watchexec -e py -- pytest                         # Run tests on save
```

Asciinema - Terminal Recorder

What is it?

Record and share terminal sessions. Perfect for documentation and demos.

Why adopt it?

- ✅ Lightweight - text-based, tiny files
- ✅ Shareable - upload or self-host
- ✅ Copy-paste enabled for viewers
- ✅ Better than video (faster, searchable, smaller)

Usage:

```
asciinema rec demo.cast                 # Start recording
# Run commands...
exit                                    # Stop recording
asciinema play demo.cast                # Replay
```

More Utilities

SSH-List - SSH connection manager with fuzzy search

```
ssh-list                    # Fuzzy search SSH config
```

CCZE - Log colorizer

```
tail -f /var/log/syslog | ccze -A    # Colorized logs
```

Makes ERROR/WARNING stand out in red/yellow!

Getting Started

Installation & Adoption Strategy

Installation

One Command Setup:

```
./setup.sh && source activate.sh && \
  ansible-playbook dev-setup.yml --ask-become-pass
```

Selective Installation (use tags):

```
ansible-playbook dev-setup.yml --tags "zsh,fzf,ripgrep"
```

All versions pinned in `vars/versions.yml`

Recommended Adoption Path

Week 1: Foundation

- Install Zsh + Oh-My-Zsh
- Start using Zellij for session management
- Adopt fzf for command history (Ctrl+R)

Week 2: Navigation & Search

- Replace ls → eza/lsd
- Use ripgrep instead of grep
- Try bat instead of cat
- Start using zoxide to jump directories

Adoption Path (continued)

Week 3: Monitoring

- Use procs instead of ps
- Install bottom for troubleshooting
- Try gping for network diagnostics

Week 4+: Specialized Tools

- Adopt gitui/lazygit for version control
- Try helix for quick edits
- Explore container tools (podman, distrobox)
- Add utilities based on your workflow

Creating Aliases for Transition

Add to `.zshrc`:

```
alias ls='eza'
alias cat='bat'
alias find='fd'
alias grep='rg'
alias ps='procs'
alias du='dust'
```

Gradual replacement of muscle memory!

Team Adoption Tips

1. **Start non-destructive** - Try tools that don't replace critical commands first
2. **Share configurations** - Commit `.zshrc` and configs to team repo
3. **Create cheatsheets** - Use Navi for team-specific commands
4. **Demo in standups** - Show one tool per week
5. **Pair sessions** - Help teammates get comfortable

Version Management:

```
python3 scripts/update-versions.py     # Check for updates
```

Conclusion

Key Takeaways

- ✅ **Productivity boost** - These tools genuinely make tasks faster and easier
- ✅ **Better UX** - Modern tools prioritize usability and helpful defaults
- ✅ **Minimal risk** - Most are drop-in replacements, easy to revert
- ✅ **Reproducible setup** - Ansible playbook = consistent environments

Next Steps

1. **Clone the repo** and run setup
2. **Start with foundation**: Zsh + Zellij + FZF
3. **Gradually adopt** tools matching your workflow
4. **Share feedback** and contribute improvements

Questions?

Let's Transform How We Work

Repository: https://github.com/jquacinella/dev-env

Documentation: Check README.md for detailed usage

Final Thought

> "These tools represent years of open-source development focused on solving real sysadmin problems.
>
> Don't try to adopt everything at once—pick 3-5 tools that address your pain points and go from there.
>
> The time investment pays back quickly."

Thank you!